

Linux Firewall and Secure Remote Access

Laboratory 4

Student Practical Laboratory Manual

Course	CORE102 TCP IP Networking and Linux Network Tools
Instructor	Aminu Idris AMCPN
Maximum score	100 marks
Gradebook category	Practical Laboratories
Estimated duration	5 to 6 hours
Platform	Kali Linux or Ubuntu Linux in an authorised laboratory environment
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Linux Firewall and Secure Remote Access – Laboratory Report

Before making any changes I snapshotted the server VM and recorded its interfaces, routes, listening sockets and running services. The Ubuntu server at 192.168.56.101 had only one network-reachable service: SSH on port 22, bound to both IPv4 and IPv6. The DNS resolver, DHCP client and NTP daemon were all local or client-side, so I kept every entry and planned to protect only port 22.

I started a temporary Python HTTP server on 8080 and confirmed the client at 192.168.56.102 could reach it. I then set UFW to default-deny incoming and default-allow outgoing, added an allow for SSH from 192.168.56.0/24, a scoped allow for 8080 from the client only, and an explicit

deny for Telnet on 23, before enabling UFW. Testing from the client confirmed SSH and 8080 were open while Telnet timed out.

I created a custom iptables chain, verified its structure and counters, then flushed and deleted it. On the client I generated an Ed25519 key pair with a passphrase, deployed the public key and confirmed public-key login. I hardened SSH via a drop-in file, validated it with `sshd -t`, reloaded the service and retested successfully. Finally I stopped the test service, removed its UFW rule, and confirmed the final socket inventory matched the baseline with no new exposure.